Supplementary Material for *Lateral Phishing as a Violation of the Organizational Communication Graph: Cascade Dynamics and Leak-Aware Evaluation*

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This supplement contains secondary tables and figures referenced from the main text. All values are produced by the reproducibility package (`code/results/*.csv`); none are new experiments.

S1. Detector hyperparameters

Model / component	Hyperparameters (fixed, without tuning)
GNN (static / TGN-lite / TGAT-lite)	DIM=24; TGAT: t_{dim}=8; 100 epochs; Adam lr=0.01, wd=$5{\cdot}10^{-4}$; dropout 0.2
GNN classifier head	MLP, 48 hidden units
Tabular detectors (LightGBM)	n_estimators=200–250, seed=42, num_leaves=31, lr=0.1
volumetric / path-based (LightGBM)	n_estimators=200, seed=42; context window W=2 buckets

Table S1: Detector hyperparameters: fixed *uniformly* for all models of a given family, framework defaults or lightly chosen, without grid search.

S2. Cascade detectors and the cascade-model sweep

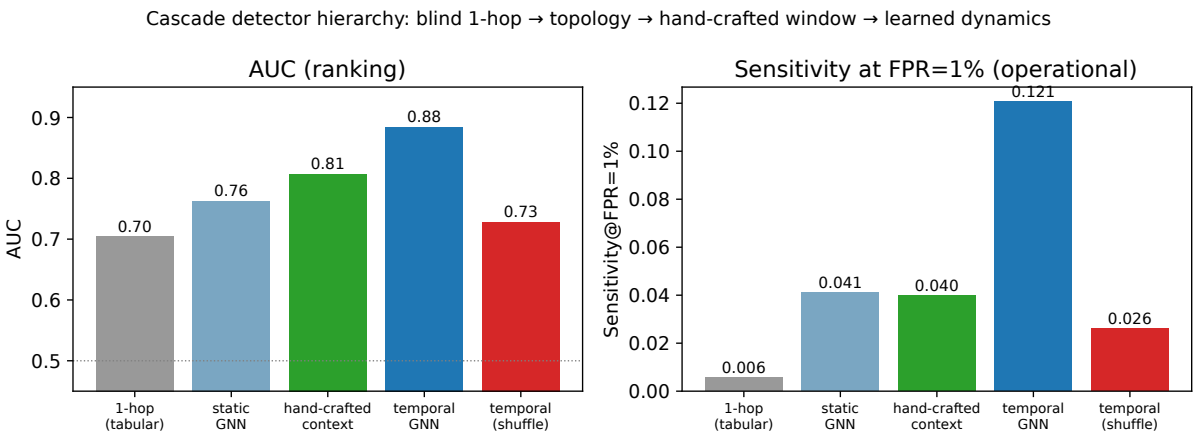


Figure S1: Detector hierarchy on AUC (left) and sensitivity@FPR1% (right); values as in the cascade-detector table of the main text. The temporal GNN dominates; time shuffling (red) levels it with the static GNN.

p_{inf}	0.05	0.1	0.2	0.3	0.5	0.7	0.9
1-hop	0.60	0.56	0.59	0.60	0.68	0.73	0.79
manual	0.75	0.77	0.77	0.79	0.79	0.81	0.85
temporal	**0.97**	**0.95**	**0.90**	**0.87**	**0.87**	**0.89**	**0.92**

Table S2: Cascade model sweep (AUC, depth 4, 8 seeds): the hierarchy holds across the whole range of the infection probability, and the receiving-side signal does not vanish at a realistically low p_{inf}.

S3. Inductive validation under increasing scale

	AUC		sensitivity@FPR1%	
Organizations	temp.	manual	temp.	manual
19	0.76	0.75	**0.062**	0.027
30	0.69	0.75	**0.041**	0.025
200	0.78	0.81	**0.076**	0.042

Table S3: Inductive validation on unseen organizations (19 organizations: 5 seeds; 30 and 200 organizations: 3 seeds): the temporal detector's operational advantage over the hand-crafted context holds on companies never seen in training at every scale (2.3×, 1.7×, 1.8×), while its AUC does not lead at the two larger scales.

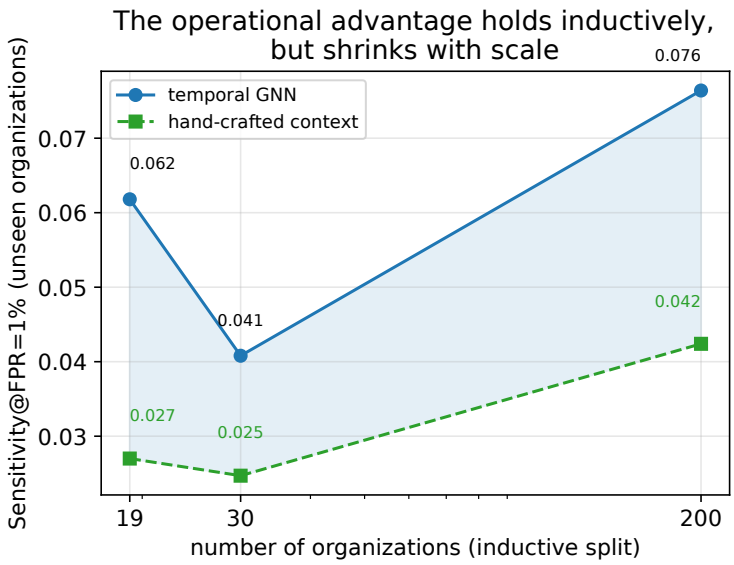


Figure S2: Inductive validation: sensitivity@FPR1% on unseen organizations against the number of companies (values as in Tab. S3). The advantage holds inductively at every scale (2.3×, 1.7×, 1.8×).

S4. Replication on real topologies

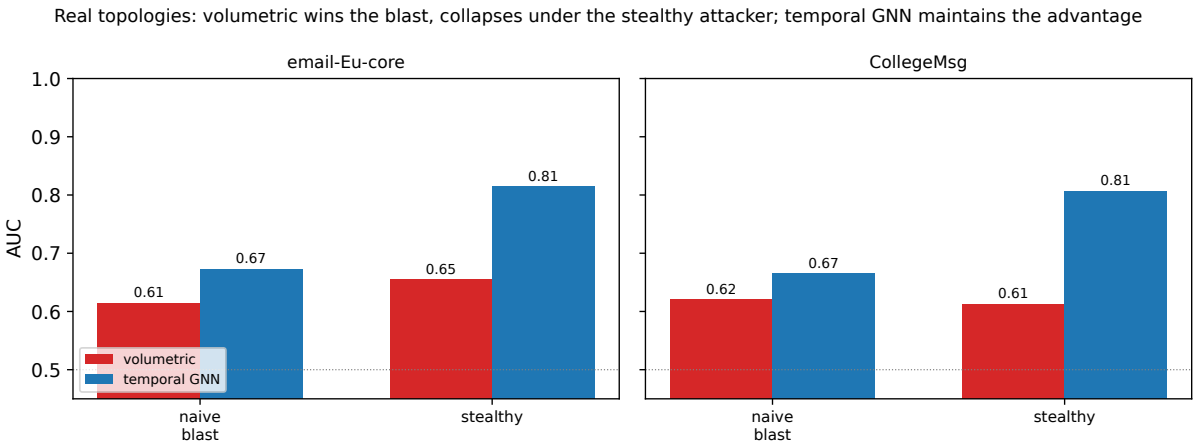


Figure S3: Two real topologies (`email-Eu-core`, `CollegeMsg`). The volumetric detector is weak in both regimes; the temporal GNN (blue) is ahead in both and its advantage is largest under the stealthy attack, where the volumetric operating point collapses. Values as in block (b) of the real-graph table of the main text.

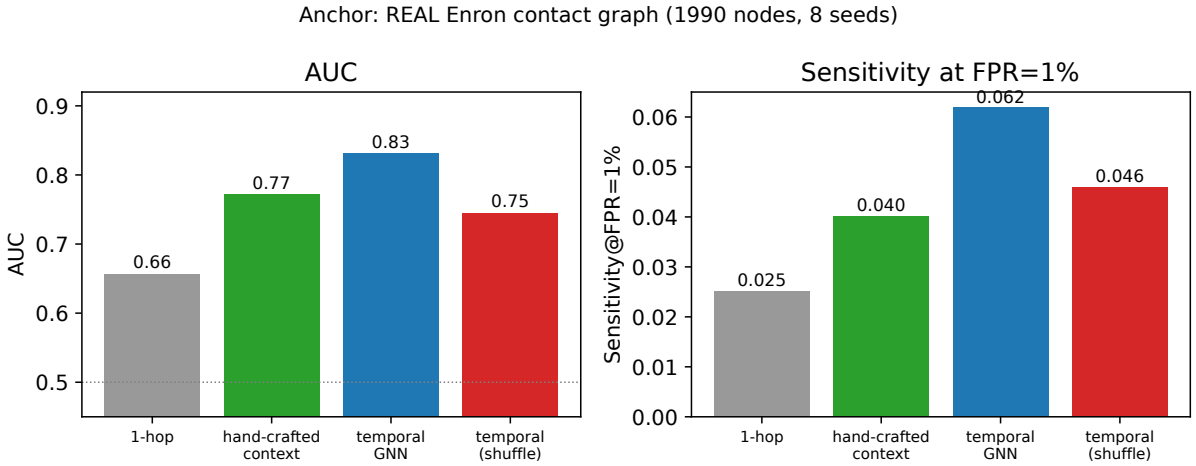


Figure S4: The cascade invariant on Enron (values as in block (a) of the real-graph table of the main text): the same hierarchy as on the case-study graph (1-hop $<$ hand-crafted context $<$ temporal GNN), with the time shuffle bringing the temporal detector back close to the hand-crafted result.

S5. Rigor: label noise and the sensitivity–FPR curve

Label noise	vol.	path	manual	temp.
0%	0.60	0.66	0.67	**0.73**
5%	0.60	0.66	0.66	**0.70**
10%	0.60	0.65	0.65	**0.66**
20%	0.59	**0.63**	0.61	0.60

Table S4: Audit of noisy training labels (AUC; stealthy attack at K=3 in the second harness (`cascadebench`, 600-account organization; the baseline table of the main text); 20 seeds): the learned temporal model degrades fastest ($0.73 \rightarrow 0.60$), so its advantage shrinks under realistic label noise.

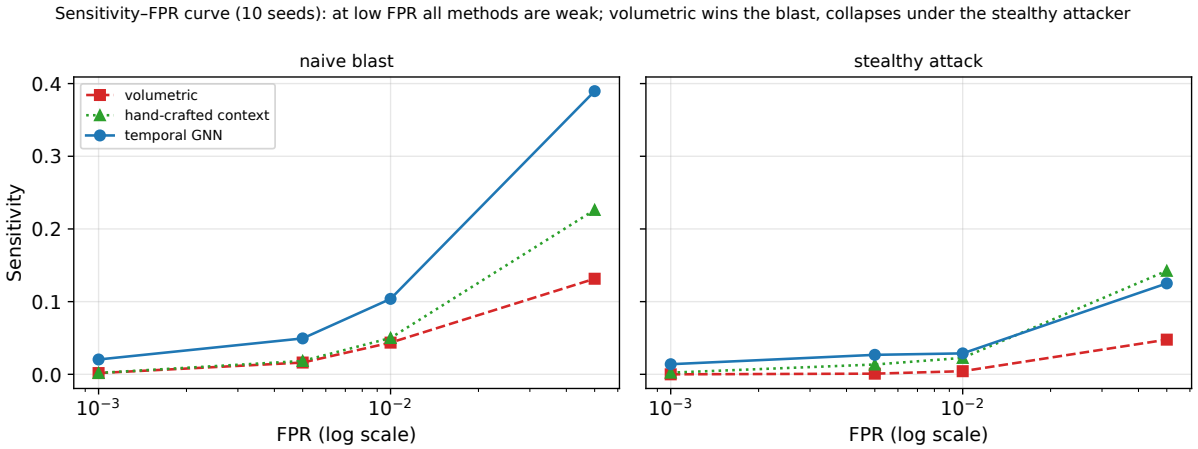


Figure S5: Sensitivity–FPR curve (10 seeds). At low FPR all methods are weak; the temporal detector stays ahead of the volumetric one in both regimes, and the gap is largest under the stealthy attack.

S6. Feature specification of every detector

All tabular detectors use the same learner (LightGBM, 200–250 trees as in Tab. S1, seed 42); they differ only in their features. Time is divided into buckets b; the context window is $W{=}2$ buckets. $\mathcal{N}(v)$ is the set of contacts of v in the communication graph; recv(b) and sent(b) are the sets of accounts that received, respectively sent, mail in bucket b; rhythm(s, b) is the temporal-consistency bit *tc*. The two experiment harnesses of the package (`experiments/` for the cascade, stealth, scale, and real-graph tables; `cascadebench/` for the baseline, label-noise, and headline-contrast tables) implement identical feature definitions.

- **1-hop (tabular).** For event $s \to v$ in bucket b: $\mathbf{1}[s \in \mathcal{N}(v)]$, rhythm$(s, b)$, $|\mathcal{N}(v)|$, $|\mathcal{N}(s)|$.
- **Hand-crafted cascade context.** The 1-hop features plus three receiving-side statistics, all computed from strictly past buckets ($\leq b{-}1$): *burst* $= |(\mathcal{N}(v) \setminus \{s\}) \cap \bigcup_{b'=b-W}^{b-1} \text{recv}(b')| / |\mathcal{N}(v)|$ (how many of the victim's other contacts were hit within the past window); *freshness* $= \mathbf{1}[s \in \bigcup_{b'=b-W}^{b-1} \text{recv}(b')]$ (the sender was itself hit just before); *propagation* $= |(\mathcal{N}(v) \setminus \{s\}) \cap \bigcup_{b'=b-W}^{b-1} \text{sent}(b')| / |\mathcal{N}(v)|$ (how many of the victim's other contacts are themselves sending).
- **Volumetric baseline (inspired by COMPA).** Three per-event deviations of the sender from its own profile, computed from strictly past buckets ($< b$): recipient novelty $1/(1 + \#\{s \to v \text{ before } b\})$, time novelty $1/(1{+}\#\{\text{events of } s \text{ in buckets} < b\})$, and sender activity $\#\{\text{events of } s \text{ before } b\}$. COMPA itself detects compromised accounts on social networks by comparing each new message with a behavioral profile of the account (time of day, source, language, links, topics); we transfer only its per-sender-deviation idea to email headers, restricted to volume and timing.
- **Path-based baseline (inspired by Hopper).** *Precursor* $= \mathbf{1}[s \in \bigcup_{b'=b-W}^{b-1} \text{recv}(b')]$ (the sender was reached just before, i.e. is part of a chain); *momentum* = number of distinct targets of s in buckets $[b{-}W, b{-}1]$; *rare target* $= 1/(1{+}\text{popularity of } v \text{ as a recipient before } b)$; *unexplained* $= 1 - \text{rhythm}(s, b)$; and the conjunction precursor $\times$ unexplained. All windows are strictly past, matching the temporal model's memory: an earlier draft of the package defined momentum over $[b, b + W]$ (future buckets) and the volumetric counters over the full stream; both were look-ahead leaks, and every experiment was re-run after the fix. Hopper itself detects lateral movement in enterprise networks from authentication logs, by reconstructing login paths between machines and flagging paths with no benign explanation; it does not process email. We transfer its causal-path idea (a precursor that explains the hop, and a rare target) to the email communication graph.
- **Static GNN.** Node features $X = \log(1{+}[\,|\mathcal{N}(u)|,\ \text{in-degree}(u),\ \text{out-degree}(u)\,])$ (three dimensions, the stream degrees computed from the *training* split only), projected to DIM=24 and propagated through two layers with the normalized adjacency (Eq. (1) of the main text). The head g is an MLP with 48 hidden units and dropout 0.2 over $[Z_s, Z_v, Z_s \odot Z_v, x_{sv}]$, where x_{sv} is the 1-hop feature vector of the event.
- **Temporal GNN (TGN-lite).** Same node features and x_{sv}; a memory of dimension 24 per node, updated bucket by bucket with a GRU cell from the mean of the messages $\sigma(W_m m_u^{(b-1)})$ of the senders that reached the node in bucket b (Eq. (3)); an event of bucket b is scored from the memories of bucket $b{-}1$ (Eq. (2)) by the same head as the static GNN. TGAT-lite replaces the mean pooling by attention with an 8-dimensional time encoding. Training: Adam (lr=0.01, weight decay $5 \cdot 10^{-4}$), 100 epochs, binary cross-entropy.

S7. Seed counts per experiment

Experiment	Table / figure	Seeds
Detector hierarchy (case study)	main-text detector table, Fig. S1	5
Library TGN (extra row)	main-text detector table	3
Architecture robustness (TGN-lite / TGAT-lite)	Sec. S1	5
Stealthy attacker sweep	main-text stealth table, Fig. S2	20
Baselines (a) stealth / (b) surgical	main-text baseline table	20
Headline contrasts (Wilcoxon, Cliff's δ)	main-text rigor section	20
Label-noise audit	Tab. S4	20
Static-blindness audit (compromised senders)	—	20
Cascade-model sweep	Tab. S2	8
Inductive validation (19 / 30 / 200 org.)	Tab. S3, Fig. S3	5 / 3 / 3
Sensitivity–FPR curve	Fig. S5	10
Enron cascade	main-text real-graph table (a), Fig. S4	8
SNAP graphs (`email-Eu-core`, `CollegeMsg`)	main-text real-graph table (b)	3

Table S5: Number of random seeds behind every reported aggregate. The stealthy, baseline, label-noise, and contrast experiments use 20 seeds; the detector hierarchy and architecture checks 5; real-graph replications use 8 (Enron) and 3 (SNAP), and are reported as directional confirmations.